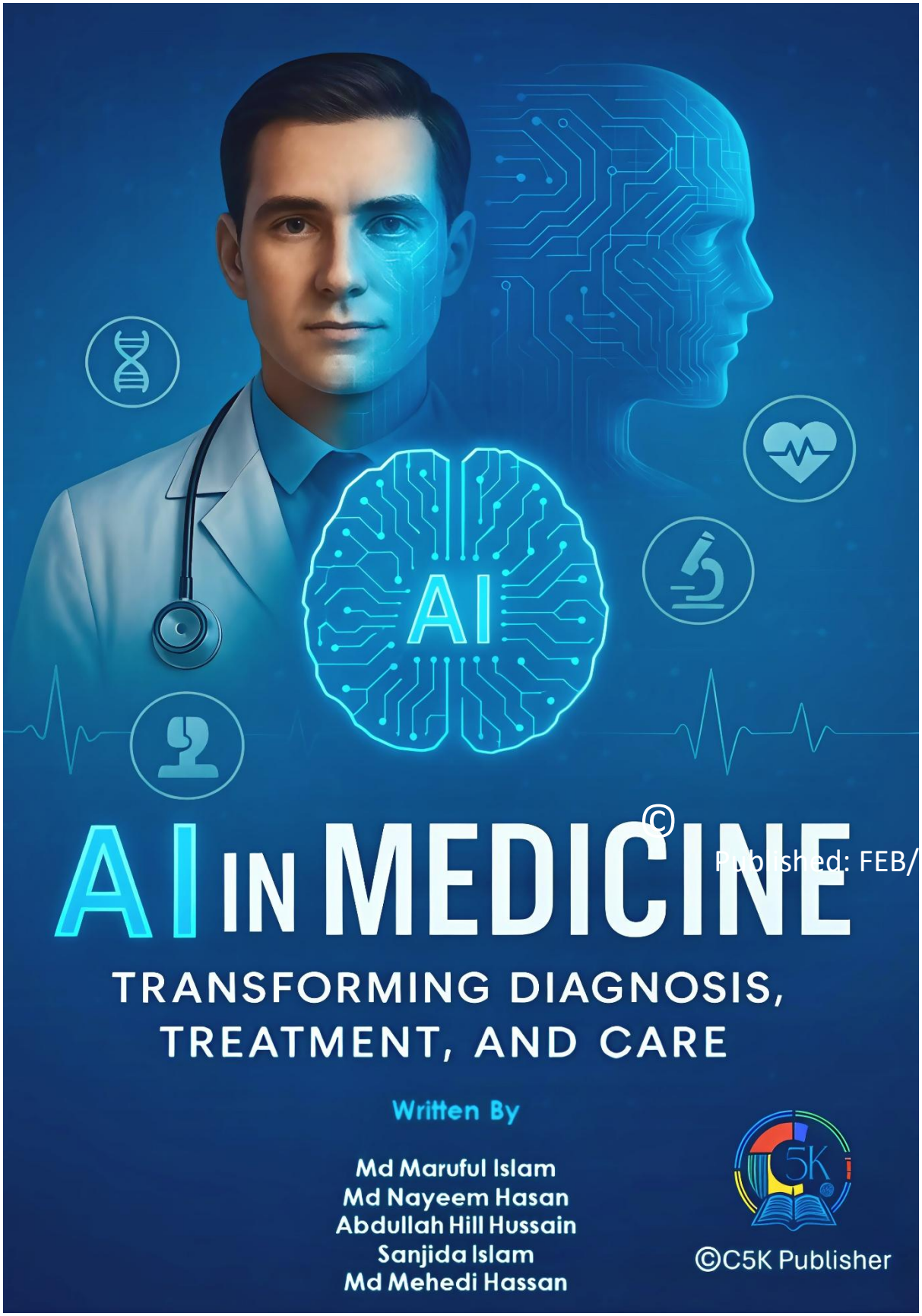

AI IN MEDICINE

TRANSFORMING DIAGNOSIS, TREATMENT, AND CARE

©
Published: FEB/

Written By

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AI in Medicine: Transforming Diagnosis, Treatment, and Care

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Authors Opinion

Artificial intelligence is ushering in one of the most profound transformations in the history of medicine. What once felt like a distant possibility is now shaping everyday clinical practice from diagnostic precision and treatment personalization to intelligent systems guiding healthcare delivery. This book was developed with a shared belief that AI is not just a tool for automation, but a powerful driver of change that redefines how we understand, manage, and improve human health. The insights presented here reflect not only cutting-edge research but also a commitment to ethical, equitable, and patient-centered innovation. As AI becomes increasingly integrated into the medical ecosystem, we must move forward with both scientific rigor and social responsibility. Sincere thanks to the entire team whose dedication, expertise, and collaboration made this work possible. Their contributions have been instrumental in shaping a comprehensive and forward-looking view of AI's role in modern healthcare. Through this work, I aimed to explore both the groundbreaking applications and the ethical crossroads we now face. AI's ability to extract insights from unstructured clinical narratives, model complex disease trajectories, and personalize treatments signals a future that is both hopeful and deeply human-centric if we design it with integrity and foresight. Yet, this evolution also demands vigilance: we must actively shape algorithms that are transparent, inclusive, and just, so that technology becomes a force for equity, not division. Ultimately, this book is a reflection of collaborative curiosity and interdisciplinary rigor. I hope it serves as a bridge between disciplines and a prompt for practitioners, researchers, and policymakers to critically engage with the promise and responsibility of AI in medicine. The future we envision depends on the decisions we make today.

-Md Maruful Islam

Contributing to AI in Medicine: Transforming Diagnosis, Treatment, and Care has been an intellectually rewarding experience. This book explores how artificial intelligence is redefining clinical decision-making, diagnostics, and patient care delivery. We've aimed to make the rapidly evolving field of AI in medicine both accessible and actionable for healthcare professionals, technologists, and policy-makers alike. In Chapter 3, we delve into machine learning and deep learning applications in clinical settings, highlighting how these tools are already assisting in early disease detection and personalized treatment planning. I hope this book empowers readers to harness the potential of AI in building more efficient and equitable healthcare systems.

-Md Nayeem Hasan

Working on this book has been a unique journey into the future of healthcare innovation. AI in Medicine captures the momentum of a digital revolution reshaping how we diagnose, monitor, and treat illness. In Chapter 5, I focused on AI-driven pathology and histopathology automation, an area where machine learning is accelerating cancer diagnostics and standardizing image analysis. Through real-world examples and research insights, we've shown how AI can enhance both speed and accuracy in clinical workflows. It's been a pleasure contributing to a book that bridges technological advancement and human-centered care.

-Abdullah Hill Hussain

Writing for this book has been a fulfilling endeavor, especially as we explored the potential of AI to close gaps in healthcare access and quality. AI in Medicine presents not just technological innovations but ethical and global perspectives on equitable care. In Chapter 21, we explore how AI can be adapted to low-resource settings, offering clinical decision support where specialists may not be available. The content resonates with a broader mission, making healthcare more inclusive and responsive to global disparities. I hope this book helps readers think globally while acting locally with AI.

-Sanjida Islam

This project has been both challenging and inspiring. Our book demystifies how AI is transforming complex systems like hospital operations and clinical support. In Chapter 13, I focused on how AI optimizes workflow and resource allocation, from intelligent scheduling to supply chain management. These innovations show how AI is not only enhancing patient outcomes but also improving institutional efficiency. I'm proud to be part of this initiative, which brings clarity to how AI is actively reshaping the behind-the-scenes mechanics of healthcare delivery.

-Md Mehedi Hassan

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